

An Eight-Charge Counterexample to the Maxwell Bound

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Abstract

We construct a configuration of eight distinct positive point charges in three-dimensional Euclidean space whose Coulomb potential has a finite critical set of cardinality at least 51, with every critical point nondegenerate. Since $51 > (8 - 1)^2 = 49$, this gives an eight-charge counterexample to the Maxwell bound. Our construction begins with a five-charge configuration having at least 24 nondegenerate critical points and adds three positive axial charges on two separated local scales. A polynomial limit accounts for 17 persistent critical points, while an innermost point-charge limit contributes 11 more. Together with the 23 nonzero critical points inherited from the five-charge seed, the total is 51.

1 Introduction

For positive strengths $q_1, \dots, q_n$ at distinct positions $a_1, \dots, a_n \in \mathbb{R}^3$, write

$$V(x) = \sum_{i=1}^n \frac{q_i}{|x - a_i|}$$

for the Coulomb potential away from the charges. A critical point is called nondegenerate if its Hessian is invertible. A configuration is called *Morse* if every critical point of its potential is nondegenerate.

In the positive-charge setting considered here, the form of Maxwell’s conjecture relevant to this paper is the following.

Maxwell’s conjecture. For every Morse configuration of n distinct positive point charges in $\mathbb{R}^3$, the number N of critical points satisfies

$$N \leq (n - 1)^2. \tag{1}$$

Some historical formulations allow signed strengths or count only isolated nondegenerate equilibria. Throughout this paper, “Maxwell’s conjecture” means precisely the positive-charge Morse statement above. The condition that the potential be Morse is the standard genericity condition in this formulation. For positive charges it also forces the critical set to be finite: every critical point lies in the convex hull of the charge positions, no critical point can approach a charge, and a compact discrete set is finite. These facts are proved directly in Section 8.

Maxwell discussed points of equilibrium in his 1873 treatise [1]. Morse and Cairns later posed the problem of finding a general upper bound [2]. Gabrielov, Novikov, and Shapiro obtained

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the first general finite bound in 2007 and, in particular, an upper bound of 12 for three charges [3]. Tsai proved the conjectured four-point upper bound for three charges of equal magnitude in 2015 [4], and Lee–Tsai exhibited nine equilibria for four positive coplanar charges in 2022 [5]. Zolotov subsequently obtained improved exponential bounds using the Thom–Milnor theorem [6]. Edelsbrunner, Fillmore, and Oliveira then proved the generic-position bound $2^n(3n-2)^3$ and constructed families whose equilibrium-to-charge ratio approaches 25/7 [7]. In 2026, Arathoon–Ball–Kvalheim produced the first counterexample to Maxwell’s conjecture [8].

We next describe precisely how the five- and seven-charge results arise. The five-charge construction of Arathoon–Ball–Kvalheim starts with three unit charges at the vertices of an equilateral triangle and adds equal charges

$$q_\epsilon = \frac{3}{4}\epsilon^3 - \frac{5}{32}\epsilon^5$$

at $\pm\epsilon e_z$. On the scale $x = \epsilon^2 X$, the central critical point of the triangle is replaced by 21 nondegenerate critical points. The three noncentral critical points of the triangle persist, which gives the lower bound

$$N_5 \geq 21 + 3 = 24 > 16.$$

Their iteration is recorded in Remark 3 and Proposition 1 of [8]. Suppose a $D_3 \times \mathbb{Z}_2$ -symmetric potential has the expansion

$$U(x) = U(0) + \alpha H_2(x) + \beta H_3(x) + O(|x|^4), \quad \alpha, \beta > 0,$$

where H_2, H_3 are defined in Section 2. Add equal charges of strength

$$q_\eta^{\text{iter}} = \alpha\eta^3 - \hat{\gamma}\eta^5$$

at $\pm\eta e_z$; this strength is positive for all sufficiently small $\eta > 0$. With

$$\lambda = \frac{2\beta}{5\alpha}, \quad \hat{\gamma} = \frac{\beta^2}{30\alpha},$$

the rescaling $x = \lambda\eta^2 X$ produces, up to a positive constant factor, the same polynomial limit used in the five-charge construction. Its 21 nondegenerate critical points persist. One replaces the former critical point at the origin, so each iteration adds two charges and gives a net gain of 20 critical points; all previously existing nonzero nondegenerate points persist. Starting from the equilateral triangle, Proposition 1 of [8] therefore gives, for every integer $m \geq 0$,

$$n = 3 + 2m, \quad N \geq 4 + 20m, \tag{2}$$

after an arbitrarily small perturbation that makes the critical set finite and entirely nondegenerate.

For $m = 1$, equation (2) is the five-charge lower bound $N_5 \geq 24$. For $m = 2$, it gives the seven-charge counterexample

$$N_7 \geq 44 > 36 = (7-1)^2.$$

It is important that Proposition 1 of [8] does not certify further violations. Its excess over Maxwell’s quantity is

$$(4 + 20m) - (2m + 2)^2 = 4m(3 - m),$$

which is strictly positive only for $m = 1, 2$. The lower bound equals Maxwell’s quantity for $m = 0$ and $m = 3$, corresponding to $n = 3$ and $n = 9$, and is smaller for $m \geq 4$. This comparison concerns only the certified lower bound; it is not an upper-bound statement about the iterated configurations.

Equivalently, Proposition 1 of [8] supplies

$$N_n \geq 10n - 26 \quad (n \geq 3 \text{ odd}).$$

Adding one sufficiently distant positive charge to the construction with $n - 1$ charges gives

$$N_n \geq 10n - 35 \quad (n \geq 6 \text{ even}).$$

For the latter lower bound,

$$(10n - 35) - (n - 1)^2 = -(n - 6)^2,$$

so this even-charge extension never proves a strict violation and reaches equality only at $n = 6$. Thus, prior to our work, Proposition 1 of [8] and its one-charge extension disproved Maxwell's conjecture only for $n = 5$ and $n = 7$. The conjecture is elementary for $n = 1, 2$, remains unresolved in full generality for $n = 3, 4, 6$, and these constructions alone do not decide it for any $n \geq 9$. Our Theorem 1 adds $n = 8$ to the list of counterexample dimensions.

The seven-charge stage is rederived below in notation adapted to the present construction. After fixing the five-charge seed W , whose expansion has positive coefficients A, B , we add a second equal pair of strength $A\eta^3 - \gamma\eta^5$ at $\pm\eta e_z$. A polynomial limit has exactly 21 nondegenerate critical points, while the 23 selected nonzero critical points of W persist, giving $23 + 21 = 44$. The eighth, much smaller, one-sided axial charge then changes only the four innermost members of that 21-point pattern: it replaces those four by 11 while preserving the other 17. The resulting count is

$$23 + (17 + 11) = 51.$$

Theorem 1 (Eight-charge counterexample). *There is a configuration of eight distinct positive point charges in $\mathbb{R}^3$ whose Coulomb potential is Morse and has a finite number $N \geq 51$ of critical points. $51 > (8 - 1)^2 = 49$, so the Maxwell bound fails for $n = 8$.*

We use the following five-charge result of Arathoon–Ball–Kvalheim [8].

Theorem 2 (Five-charge seed). *There is an $\epsilon_0 > 0$ such that, for every $0 < \epsilon < \epsilon_0$, one may place three unit charges at the vertices of an equilateral triangle of circumradius one and equal charges*

$$q_\epsilon = \frac{3}{4}\epsilon^3 - \frac{5}{32}\epsilon^5$$

at $\pm\epsilon e_z$. The resulting Coulomb potential has at least 24 nondegenerate critical points. Consequently, at least 23 of them are nonzero. The result does not exclude additional critical points.

The word “nonzero” in the preceding theorem refers to the origin in the coordinates of the equilateral triangle. In Section 2 we verify directly that the origin is one critical point of the five-charge potential; hence a set of at least 24 distinct nondegenerate critical points contains at least 23 points away from the origin. We fix any such 23 points for the persistence argument below.

The remainder of the paper constructs the three additional charges and proves Theorem 1. Section 2 establishes the local expansion of the imported seed. Sections 3–4 define the three new charges and obtain the first local limit. Sections 5–7 prove the $17 + 11$ local count and combine it with the 23 persistent seed points. Section 8 removes any remaining degeneracies and proves finiteness.

2 Notation and the five-charge seed

Let $e_z = (0, 0, 1)$. In Cartesian coordinates (x, y, z) , put $\rho^2 = x^2 + y^2$; when $\rho > 0$, let θ denote the azimuthal angle, so that $x = \rho \cos \theta$ and $y = \rho \sin \theta$. Define

$$H_2 = \rho^2 - 2z^2, \quad H_3 = x^3 - 3xy^2 = \rho^3 \cos(3\theta),$$

and

$$H_4 = 3\rho^4 - 24\rho^2 z^2 + 8z^4.$$

These are homogeneous harmonic polynomials of degrees 2, 3, and 4, respectively. The cylindrical expression for H_3 is used only when $\rho > 0$; its Cartesian expression defines it everywhere.

The triangle vertices are

$$a_1 = (1, 0, 0), \quad a_2 = (-1/2, \sqrt{3}/2, 0), \quad a_3 = (-1/2, -\sqrt{3}/2, 0).$$

They satisfy $|a_j| = 1$ and $|a_j - a_k| = \sqrt{3}$ for $j \neq k$. Thus the triangle is equilateral with circumradius one (and side length $\sqrt{3}$).

Choose $\epsilon > 0$ sufficiently small for Theorem 2, shrinking it if necessary so that $\epsilon < 1$. Since

$$q_\epsilon = \epsilon^3 \left(\frac{3}{4} - \frac{5}{32}\epsilon^2 \right),$$

the quantity in parentheses is at least $19/32$, and hence the axial strength q_ϵ is positive. Let W denote the five-charge potential

$$W(x) = \sum_{j=1}^3 \frac{1}{|x - a_j|} + \frac{q_\epsilon}{|x - \epsilon e_z|} + \frac{q_\epsilon}{|x + \epsilon e_z|},$$

defined on $\mathbb{R}^3 \setminus \{a_1, a_2, a_3, \epsilon e_z, -\epsilon e_z\}$.

Proposition 3 (Expansion of the seed potential). *The potential W is real analytic near the origin and has the expansion*

$$W(x) = W(0) + AH_2(x) + BH_3(x) + O_\epsilon(|x|^4), \quad (3)$$

where

$$A = \frac{5}{32}\epsilon^2 > 0, \quad B = \frac{15}{8} > 0. \quad (4)$$

More precisely,

$$W(x) = W(0) + \frac{5}{32}\epsilon^2 H_2(x) + \frac{15}{8} H_3(x) + \left(\frac{3}{16\epsilon^2} + \frac{13}{128} \right) H_4(x) + O_\epsilon(|x|^5),$$

where $W(0) = 3 + 2q_\epsilon/\epsilon$. If $\mathcal{E}_\epsilon$ denotes the remainder after the constant, quadratic, and cubic terms in equation (3), then for every multi-index α with $|\alpha| \leq 2$, there are constants $r_\epsilon, C_\epsilon > 0$ such that

$$|D^\alpha \mathcal{E}_\epsilon(x)| \leq C_\epsilon |x|^{4-|\alpha|} \quad (|x| < r_\epsilon).$$

Proof. Write $V_\Delta(x) = \sum_{j=1}^3 |x - a_j|^{-1}$ for the potential of the three unit charges. For any unit vector a , the Taylor expansion at the origin is

$$\begin{aligned} \frac{1}{|x - a|} &= 1 + a \cdot x + \frac{1}{2}(3(a \cdot x)^2 - |x|^2) \\ &\quad + \frac{1}{2}(5(a \cdot x)^3 - 3|x|^2(a \cdot x)) \\ &\quad + \frac{1}{8}(35(a \cdot x)^4 - 30|x|^2(a \cdot x)^2 + 3|x|^4) + O(|x|^5). \end{aligned}$$

For the three vectors a_j , direct calculation gives

$$\sum_{j=1}^3 a_j = 0, \quad \sum_{j=1}^3 (a_j \cdot x)^2 = \frac{3}{2}\rho^2, \quad \sum_{j=1}^3 (a_j \cdot x)^3 = \frac{3}{4}H_3(x),$$

and

$$\sum_{j=1}^3 (a_j \cdot x)^4 = \frac{9}{8} \rho^4.$$

Substitution into the preceding Taylor formula yields

$$V_\Delta(x) = 3 + \frac{3}{4}H_2(x) + \frac{15}{8}H_3(x) + \frac{9}{64}H_4(x) + O(|x|^5).$$

For $|x| < \epsilon$, the generating function for the Legendre polynomials, or direct Taylor expansion, gives the even expansion of the axial pair:

$$\frac{q_\epsilon}{|x - \epsilon e_z|} + \frac{q_\epsilon}{|x + \epsilon e_z|} = \frac{2q_\epsilon}{\epsilon} - \frac{q_\epsilon}{\epsilon^3}H_2(x) + \frac{q_\epsilon}{4\epsilon^5}H_4(x) + O\left(\frac{q_\epsilon|x|^6}{\epsilon^7}\right).$$

In particular, the axial pair contributes no odd-degree terms. Using the definition of q_ϵ , the combined quadratic coefficient is

$$\frac{3}{4} - \frac{q_\epsilon}{\epsilon^3} = \frac{3}{4} - \left(\frac{3}{4} - \frac{5}{32}\epsilon^2\right) = \frac{5}{32}\epsilon^2,$$

whereas the cubic coefficient remains $15/8$. The combined quartic coefficient is

$$\frac{9}{64} + \frac{q_\epsilon}{4\epsilon^5} = \frac{9}{64} + \frac{3}{16\epsilon^2} - \frac{5}{128} = \frac{3}{16\epsilon^2} + \frac{13}{128}.$$

This proves both displayed expansions and the values in equation (4).

Finally, every charge is a positive distance from the origin, so W is real analytic there. Its Taylor series after subtraction of the terms of degree at most three begins in degree four. Taylor's theorem applied also to derivatives of order at most two therefore gives the asserted bounds on $D^\alpha \mathcal{E}_\epsilon$. A direct calculation also confirms $\Delta H_j = 0$ for $j = 2, 3, 4$, completing all claims. $\square$

In particular, the origin is a nondegenerate critical point of W , since

$$D^2W(0) = \text{diag}(2A, 2A, -4A).$$

Consequently, for every compact set $\mathcal{K} \subset \mathbb{R}^3$, the chain rule and the derivative estimate in Proposition 3 give

$$\left\| \eta^{-6} \mathcal{E}_\epsilon(\eta^2 \cdot) \right\| * C^2(\mathcal{K}) = O * \epsilon, \mathcal{K}(\eta^2) \quad \text{as } \eta \downarrow 0.$$

In every subsequent limit, ϵ is fixed first. Thus the constants implicit in $O_\epsilon(\cdot)$ and in the derivative estimates above need not be uniform as $\epsilon \downarrow 0$.

3 The eight charges

Fix

$$\kappa = 10, \quad \mu = \frac{1}{2}.$$

Choose $\gamma > 0$ sufficiently small, and then choose $\eta > 0$ sufficiently small. Add the following three charges to the five-charge configuration defining W :

1. Equal charges

$$p_{\gamma, \eta} = A\eta^3 - \gamma\eta^5 \tag{5}$$

at $\pm \eta e_z$.

2. One charge

$$Q_{\gamma,\eta} = \frac{2048}{27}\gamma^4\eta^8 \quad (6)$$

at δe_z , where

$$\delta = \frac{16}{3}\gamma\eta^2. \quad (7)$$

Denote the resulting eight-charge potential by

$$V_{\gamma,\eta}(x) = W(x) + \frac{p_{\gamma,\eta}}{|x - \eta e_z|} + \frac{p_{\gamma,\eta}}{|x + \eta e_z|} + \frac{Q_{\gamma,\eta}}{|x - \delta e_z|}.$$

Here ϵ has already been fixed in Section 2. The parameter γ will be fixed sufficiently small in Sections 5–7; only then is η chosen. For fixed $\gamma > 0$,

$$p_{\gamma,\eta} = \eta^3(A - \gamma\eta^2) > 0 \quad \text{if} \quad \eta^2 < \frac{A}{\gamma},$$

and $\delta/\eta = (16/3)\gamma\eta \rightarrow 0$ as $\eta \downarrow 0$. Hence η can be chosen so that $p_{\gamma,\eta} > 0$ and $0 < \delta < \eta < \epsilon$. The eight positions are then distinct and all eight strengths are positive.

4 The first local limit

For an axial pair of strength p at $\pm\eta e_z$, Taylor expansion at the origin gives

$$\frac{p}{|x - \eta e_z|} + \frac{p}{|x + \eta e_z|} = \frac{2p}{\eta} - \frac{p}{\eta^3}H_2(x) + \frac{p}{4\eta^5}H_4(x) + O\left(\frac{p|x|^6}{\eta^7}\right). \quad (8)$$

As in Proposition 3, the remainder in equation (8) may be differentiated twice, with the corresponding loss of powers of $|x|$. On a fixed compact set in the scaled variable X , the substitution $x = \eta^2 X$ satisfies $|x| < \eta$ for all sufficiently small η , so this expansion applies uniformly on that compact set.

Set

$$\tilde{V}_{\gamma,\eta}(X) = \frac{V_{\gamma,\eta}(\eta^2 X) - V_{\gamma,\eta}(0)}{\eta^6}.$$

The candidate limit is

$$K_\gamma(X) = P_\gamma(X) + \frac{(2048/27)\gamma^4}{|X - (16\gamma/3)e_z|}, \quad (9)$$

where

$$P_\gamma = \gamma H_2 + B H_3 + \frac{A}{4} H_4. \quad (10)$$

Proposition 4 (First local limit). *Fix $\gamma > 0$. For every compact set*

$$\mathcal{K} \subset \mathbb{R}^3 \setminus \left\{ \frac{16\gamma}{3} e_z \right\},$$

one has

$$\left\| \tilde{V}_{\gamma,\eta} - (K_\gamma - K_\gamma(0)) \right\| * C^2(\mathcal{K}) = O * \epsilon, \gamma, \mathcal{K}(\eta^2) \quad \text{as } \eta \downarrow 0.$$

Proof. By homogeneity and Proposition 3,

$$\frac{W(\eta^2 X) - W(0)}{\eta^6} = A\eta^{-2}H_2(X) + B H_3(X) + \mathcal{E}^W * \eta(X), \quad \|\mathcal{E}^W * \eta\| * C^2(\mathcal{K}) = O * \epsilon, \mathcal{K}(\eta^2).$$

The nonconstant part of the second axial pair, after the same substitution and normalization, is

$$-(A\eta^{-2} - \gamma)H_2(X) + \left(\frac{A}{4} - \frac{\gamma\eta^2}{4}\right)H_4(X) + \mathcal{E}^{\text{pair}} * \eta(X),$$

where

$$\|\mathcal{E}^{\text{pair}} * \eta\| * C^2(\mathcal{K}) = O * \epsilon, \gamma, \mathcal{K}(\eta^2).$$

Thus the terms $A\eta^{-2}H_2$ cancel exactly and the polynomial terms that remain are P_γ , up to a C^2 -error of order η^2 .

Finally, equations (6) and (7) give the exact identity

$$\frac{Q_{\gamma,\eta}}{\eta^6} \left(\frac{1}{|\eta^2 X - \delta e_z|} - \frac{1}{|\delta e_z|} \right) = \frac{2048}{27} \gamma^4 \left(\frac{1}{|X - (16\gamma/3)e_z|} - \frac{1}{|(16\gamma/3)e_z|} \right).$$

Combining the three contributions proves the stated estimate. $\square$

5 Critical points of the polynomial part

This section and Section 6 both compute Hessians in cylindrical coordinates, so we first record the two elementary facts that will be used in each.

Let f be smooth near a point with $\rho > 0$, and let (e_ρ, e_θ, e_z) be the orthonormal cylindrical frame there. The entries of the Hessian bilinear form D^2f in this frame are

$$\begin{aligned} D^2f(e_\rho, e_\rho) &= \partial_{\rho\rho}f, & D^2f(e_\theta, e_\theta) &= \rho^{-2}\partial_{\theta\theta}f + \rho^{-1}\partial_\rho f, & D^2f(e_z, e_z) &= \partial_{zz}f, \\ D^2f(e_\rho, e_z) &= \partial_{\rho z}f, & D^2f(e_\rho, e_\theta) &= \rho^{-1}\partial_{\rho\theta}f - \rho^{-2}\partial_\theta f, & D^2f(e_\theta, e_z) &= \rho^{-1}\partial_{\theta z}f. \end{aligned} \quad (11)$$

Since the frame is orthonormal, this matrix is the Cartesian Hessian written in a rotated basis; its eigenvalues, determinant, and signature are therefore those of D^2f .

Lemma 5 (Angular decoupling). *Let p be a critical point of f with $\rho > 0$, and suppose $\partial_\rho f(p) = \partial_{\theta z}f(p) = 0$. Then the matrix (11) is block diagonal at p , with meridional block*

$$D_{\rho,z}^2 f = \begin{pmatrix} \partial_{\rho\rho}f & \partial_{\rho z}f \\ \partial_{\rho z}f & \partial_{zz}f \end{pmatrix}$$

and a single angular eigenvalue $\rho^{-2}\partial_{\theta\theta}f$. In particular $\det D^2f = \rho^{-2}\partial_{\theta\theta}f \cdot \det D_{\rho,z}^2 f$.

Proof. At a critical point $\partial_\rho f = \partial_\theta f = 0$, so both correction terms in equation (11) vanish; the hypothesis annihilates the two remaining off-block entries. $\square$

Every function to which Lemma 5 is applied below has the form $u(\rho, z) + c\rho^3 \cos(3\theta)$. For such a function $\partial_{\theta z}f \equiv 0$ and $\partial_{\rho\theta}f = -3c\rho^2 \sin(3\theta)$, which vanishes at every critical point with $\rho > 0$ because the equation $\partial_\theta f = -3c\rho^3 \sin(3\theta) = 0$ forces $\sin(3\theta) = 0$. The hypothesis of Lemma 5 is therefore automatic.

Lemma 6 (Saddle dichotomy). *Let f be harmonic on an open subset of $\mathbb{R}^3$ and let p be a nondegenerate critical point of f . Then the Morse index of p is 1 or 2; it is 1 when $\det D^2f(p) < 0$ and 2 when $\det D^2f(p) > 0$.*

Proof. The Hessian is symmetric with vanishing trace and no zero eigenvalue, so its three real eigenvalues are nonzero and sum to zero. Hence at least one is positive and at least one is negative, so the number k of negative eigenvalues is 1 or 2. The determinant is the product of the eigenvalues, so its sign is $(-1)^k$. $\square$

The polynomial P_γ is a linear combination of H_2 , H_3 , and H_4 , hence harmonic, so Lemma 6 applies to it.

Proposition 7. *Assume*

$$\frac{A\gamma}{B^2} < \frac{3}{80}. \quad (12)$$

Then P_γ has exactly 21 critical points, all nondegenerate: ten of Morse index 1 and eleven of Morse index 2. As $\gamma \downarrow 0$, four lie in an $O(\gamma)$ -neighborhood of the origin (the origin itself, of index 1, together with three points of index 2 and of norm asymptotic to $2\gamma/(3B)$); eight have norm $\Theta(\sqrt{\gamma})$; and nine converge to distinct nonzero limits. The eight points of norm $\Theta(\sqrt{\gamma})$ have mutual distances $\Theta(\gamma)$, not $\Theta(\sqrt{\gamma})$.

Proof. In cylindrical coordinates,

$$P_\gamma = \gamma(\rho^2 - 2z^2) + B\rho^3 \cos(3\theta) + \frac{A}{4}(3\rho^4 - 24\rho^2 z^2 + 8z^4).$$

Its first derivatives are

$$\begin{aligned} \partial_\rho P_\gamma &= \rho(2\gamma + 3B\rho \cos(3\theta) + 3A\rho^2 - 12Az^2), \\ \partial_\theta P_\gamma &= -3B\rho^3 \sin(3\theta), \\ \partial_z P_\gamma &= 4z(-\gamma - 3A\rho^2 + 2Az^2). \end{aligned}$$

When $\rho = 0$, the Cartesian critical equations reduce to

$$4z(-\gamma + 2Az^2) = 0,$$

giving the origin and the two axial points listed below. Suppose henceforth that $\rho > 0$. Then $\sin(3\theta) = 0$, so put $s = \cos(3\theta) \in \{+1, -1\}$. If $z = 0$, the radial equation is

$$2\gamma + 3Bs\rho + 3A\rho^2 = 0.$$

It has positive solutions only when $s = -1$, in which case it is equation (13). If $z \neq 0$, the vertical equation gives

$$z^2 = \frac{\gamma}{2A} + \frac{3}{2}\rho^2.$$

Substitution in the radial equation gives

$$15A\rho^2 - 3Bs\rho + 4\gamma = 0.$$

This has positive solutions only when $s = +1$, and then it is equation (14). Thus the following list is complete.

1. The origin.
2. Two axial points:

$$\rho = 0, \quad z = \pm \sqrt{\frac{\gamma}{2A}}.$$

3. Six planar points. Here $z = 0$, $\cos(3\theta) = -1$, and ρ is either positive root of

$$3A\rho^2 - 3B\rho + 2\gamma = 0. \quad (13)$$

Each radius supplies three angles.

4. Twelve off-planar points. Here $\cos(3\theta) = 1$, ρ is either positive root of

$$15A\rho^2 - 3B\rho + 4\gamma = 0, \quad (14)$$

and

$$z = \pm \sqrt{\frac{\gamma}{2A} + \frac{3}{2}\rho^2}.$$

Each radius supplies three angles and two signs of z .

Condition (12) makes the discriminant in equation (14) positive; it also implies positivity of the discriminant in equation (13). Since the sum and product of the roots in each equation are positive, all four roots are positive.

We now verify nondegeneracy. At the origin,

$$D^2P_\gamma(0) = \text{diag}(2\gamma, 2\gamma, -4\gamma).$$

At either axial point, the three Hessian eigenvalues are

$$(-4\gamma, -4\gamma, 8\gamma).$$

At a point with $\rho > 0$ and $\sin(3\theta) = 0$, Lemma 5 applies, and the angular eigenvalue is

$$\rho^{-2}\partial_{\theta\theta}P_\gamma = -9Bs\rho \neq 0.$$

At a planar point $s = -1$ and $z = 0$. There the mixed entry vanishes,

$$\partial_{\rho z}P_\gamma = \partial_\rho(4z(-\gamma - 3A\rho^2 + 2Az^2)) = -24A\rho z = 0,$$

so the meridional block is itself diagonal, and its two entries, after using equation (13), are the radial and vertical eigenvalues

$$3\rho(2A\rho - B) = \rho \frac{d}{d\rho}(3A\rho^2 - 3B\rho + 2\gamma) \quad \text{and} \quad -4\gamma - 12A\rho^2.$$

The second is negative, and the first is nonzero because the two roots of equation (13) are simple. At an off-planar point the determinant of the meridional (ρ, z) -block is

$$48A\rho z^2(B - 10A\rho), \quad (15)$$

which is nonzero because

$$\frac{d}{d\rho}(15A\rho^2 - 3B\rho + 4\gamma) = -3(B - 10A\rho)$$

does not vanish at a simple root. The angular eigenvalue is also nonzero. Thus all 21 listed points are nondegenerate.

We next determine the Morse indices. Since P_γ is harmonic, Lemma 6 reduces this to the sign of $\det D^2P_\gamma$, which for $\rho > 0$ factors as in Lemma 5. At the origin the determinant is $-16\gamma^3 < 0$ and at an axial point it is $128\gamma^3 > 0$, so the origin has index 1 and the two axial points index 2. At a planar point the angular eigenvalue $9B\rho$ is positive and the vertical eigenvalue is negative, so the sign of the determinant is opposite to that of the radial eigenvalue, hence opposite to that of $\frac{d}{d\rho}(3A\rho^2 - 3B\rho + 2\gamma)$. This derivative is negative at the smaller root $\rho_{p,-}$ and positive at the larger root $\rho_{p,+}$, so the three points of radius $\rho_{p,-}$ have index 2 and the three of radius $\rho_{p,+}$ have index 1. At an off-planar point the angular eigenvalue $-9B\rho$ is negative, so the sign of the determinant is opposite to that of equation (15), hence opposite to that of $B - 10A\rho$. The displayed derivative shows that $B - 10A\rho > 0$ at the smaller root $\rho_{o,-}$ of equation (14) and

$B - 10A\rho < 0$ at the larger root $\rho_{o,+}$, so the six points of radius $\rho_{o,-}$ have index 1 and the six of radius $\rho_{o,+}$ have index 2. In total

$$m_1 = 1 + 3 + 6 = 10, \quad m_2 = 2 + 3 + 6 = 11. \quad (16)$$

We turn to the asymptotics. Recall that $\rho_{p,-} < \rho_{p,+}$ denote the roots of equation (13) and $\rho_{o,-} < \rho_{o,+}$ those of equation (14). Expanding the quadratic formulas gives

$$\begin{aligned} \rho_{p,-} &= \frac{2\gamma}{3B} + O(\gamma^2), & \rho_{p,+} &= \frac{B}{A} - \frac{2\gamma}{3B} + O(\gamma^2), \\ \rho_{o,-} &= \frac{4\gamma}{3B} + O(\gamma^2), & \rho_{o,+} &= \frac{B}{5A} - \frac{4\gamma}{3B} + O(\gamma^2). \end{aligned}$$

The origin and the three points with radius $\rho_{p,-}$ therefore lie in an $O(\gamma)$ -neighborhood of the origin; the three nonzero points have norm asymptotic to $2\gamma/(3B)$. The two axial points and the six off-planar points with radius $\rho_{o,-}$ have norm $\Theta(\sqrt{\gamma})$, because for the latter

$$z^2 = \frac{\gamma}{2A} + O(\gamma^2).$$

The remaining three planar and six off-planar points converge to distinct nonzero limits. This proves the asserted decomposition $4 + 8 + 9 = 21$.

It remains to record the mutual separations of the eight points of norm $\Theta(\sqrt{\gamma})$; these are used in Section 7, and the fact that they are *smaller* than the norms is what dictates the scale of the balls employed there. The eight points are the two axial points $(0, 0, \pm\sqrt{\gamma/(2A)})$ and the six off-planar points of radius $\rho_{o,-} = 4\gamma/(3B) + O(\gamma^2)$ and heights $z = \pm\sqrt{\gamma/(2A)} + O(\gamma^2)$. Thus, although each of the eight has norm $\Theta(\sqrt{\gamma})$, the six off-planar ones hug the axis, lying at horizontal distance only $\rho_{o,-} = \Theta(\gamma)$ from it. The vertical gap between an off-planar point and the axial point with the same sign of z is

$$\sqrt{\frac{\gamma}{2A} + \frac{3}{2}\rho_{o,-}^2} - \sqrt{\frac{\gamma}{2A}} = O(\gamma^{3/2}),$$

so their distance is $(\rho_{o,-}^2 + O(\gamma^3))^{1/2} = \rho_{o,-}(1 + O(\gamma)) = \Theta(\gamma)$. Two off-planar points with the same sign of z are at distance $\sqrt{3}\rho_{o,-} = \Theta(\gamma)$, and every remaining pair is separated by a vertical distance $2\sqrt{\gamma/(2A)} + O(\gamma^{3/2}) = \Theta(\sqrt{\gamma})$. The minimum mutual distance among the eight is therefore $\Theta(\gamma)$. $\square$

6 Eleven points on the innermost scale

Put

$$d = \frac{16}{3}\gamma = \frac{\kappa\gamma}{B}, \quad X = dY.$$

The constants in Section 3 were chosen so that

$$\frac{Bd}{\gamma} = \kappa = 10, \quad \frac{(2048/27)\gamma^4}{\gamma d^3} = \mu = \frac{1}{2}.$$

By homogeneity, division of equation (9) by γd^2 gives the exact identity

$$\frac{K_\gamma(dY)}{\gamma d^2} = H_2(Y) + 10H_3(Y) + \frac{64A\gamma}{9}H_4(Y) + \frac{1}{2|Y - e_z|}. \quad (17)$$

It therefore converges in C^2 , away from e_z , to

$$F(Y) = H_2(Y) + 10H_3(Y) + \frac{1}{2|Y - e_z|}. \quad (18)$$

The function F is harmonic on $\mathbb{R}^3 \setminus \{e_z\}$, being the sum of two harmonic polynomials and a Newtonian kernel, so Lemma 6 applies to it.

Lemma 8. *The function F has exactly 11 critical points, all nondegenerate: four of Morse index 1 and seven of Morse index 2.*

Proof. Write

$$\mathcal{R} = \sqrt{\rho^2 + (z - 1)^2}.$$

In cylindrical coordinates the first derivatives of F are

$$\begin{aligned} \partial_\rho F &= 2\rho + 30\rho^2 \cos(3\theta) - \frac{\rho}{2\mathcal{R}^3}, \\ \partial_\theta F &= -30\rho^3 \sin(3\theta), \\ \partial_z F &= -4z - \frac{z-1}{2\mathcal{R}^3}. \end{aligned}$$

First suppose $\rho = 0$. There is no solution with $z > 1$, because both terms in $\partial_z F$ are then negative. For $z < 1$, the axial critical equation reduces to

$$z(1-z)^2 = \frac{1}{8}, \quad 0 < z < 1.$$

Here the restriction $z > 0$ follows from the equation itself. The cubic factors as

$$8z(1-z)^2 - 1 = (2z-1)(4z^2 - 6z + 1).$$

Its roots are $1/2$ and $(3 \pm \sqrt{5})/4$; the root with the plus sign is larger than one. Thus the two admissible roots are

$$z = \frac{1}{2}, \quad z = \frac{3 - \sqrt{5}}{4}. \quad (19)$$

At an axial critical point the Hessian is diagonal. Indeed, in Cartesian coordinates $F = (x^2 + y^2 - 2z^2) + 10(x^3 - 3xy^2) + \frac{1}{2}\mathcal{R}^{-1}$; the Hessian of the cubic term is $60\begin{pmatrix} x & -y & 0 \\ -y & -x & 0 \\ 0 & 0 & 0 \end{pmatrix}$, which vanishes at $x = y = 0$, while the two remaining terms are invariant under rotation about the z -axis and hence have Hessians diagonal on that axis. The repeated transverse eigenvalue and the axial eigenvalue are, using $1/(2(1-z)^2) = 4z$ from the critical equation,

$$\lambda_\perp = 2 - \frac{1}{2(1-z)^3} = \frac{2(1-3z)}{1-z}, \quad \lambda_z = -4 + \frac{1}{(1-z)^3} = \frac{4(3z-1)}{1-z}.$$

Neither root in equation (19) equals $1/3$, so both axial critical points are nondegenerate. Moreover $\det D^2 F = \lambda_\perp^2 \lambda_z$ has the sign of $3z-1$; since $\frac{1}{2} > \frac{1}{3} > (3 - \sqrt{5})/4$, Lemma 6 gives index 2 at $z = 1/2$ and index 1 at $z = (3 - \sqrt{5})/4$.

Now suppose $\rho > 0$. The angular equation gives $s = \cos(3\theta) \in \{+1, -1\}$. The radial equation is

$$\frac{1}{2\mathcal{R}^3} = 2 + 30s\rho,$$

so admissibility requires $2 + 30s\rho > 0$. Substituting this identity into the vertical equation and solving for z gives

$$z = \frac{2 + 30s\rho}{6 + 30s\rho} \quad (20)$$

and hence $1 - z = 4/(6 + 30s\rho)$. The radial equation is therefore equivalent to

$$C_s(\rho) := (2 + 30s\rho) \left(\rho^2 + \frac{16}{(6 + 30s\rho)^2} \right)^{3/2} = \frac{1}{2}. \quad (21)$$

For $s = +1$, put $h = C_+^2$. Direct differentiation gives

$$h'(\rho) = \frac{8\rho(15\rho + 1)(225\rho^4 + 90\rho^3 + 9\rho^2 + 4)^2 \Xi(\rho)}{243(5\rho + 1)^7}, \quad (22)$$

where

$$\Xi(\rho) = 22500\rho^4 + 14625\rho^3 + 3375\rho^2 + 315\rho - 191.$$

One has

$$\Xi'(\rho) = 90000\rho^3 + 43875\rho^2 + 6750\rho + 315 > 0 \quad (\rho > 0),$$

while $\Xi(0) = -191$ and $\Xi(\rho) \rightarrow +\infty$. Thus the polynomial Ξ has exactly one positive root, which we call ρ_* ; it is negative on $(0, \rho_*)$ and positive on (ρ_*, ∞) . All other factors in equation (22) are positive for $\rho > 0$. Hence h , and therefore the positive function $C_+ = \sqrt{h}$, is strictly decreasing on $(0, \rho_*)$ and strictly increasing on (ρ_*, ∞) . Exact substitution gives

$$C_+(0) > \frac{1}{2}, \quad C_+(1/10) < \frac{1}{2}, \quad C_+(1/4) > \frac{1}{2}. \quad (23)$$

The corresponding squared values are

$$\frac{256}{729}, \quad \frac{4750104241}{21257640000}, \quad \frac{1944701504497}{6347497291776}.$$

The first and third fractions are greater than $1/4$, while the middle fraction is less than $1/4$; as $C_+ > 0$, the comparisons (23) follow.

We now count the solutions of $C_+ = 1/2$ on $(0, \infty)$. Because ρ_* is the global minimizer, $C_+(\rho_*) \leq C_+(1/10) < 1/2$; no case distinction according to the position of ρ_* relative to $1/10$ is needed. On $(0, \rho_*]$ the function C_+ decreases strictly from $C_+(0) = 16/27 > 1/2$ to $C_+(\rho_*) < 1/2$, so it attains the value $1/2$ exactly once there. On $[\rho_*, \infty)$ it increases strictly from $C_+(\rho_*) < 1/2$ to $+\infty$, since $C_+(\rho) \sim 30\rho^4$ as $\rho \rightarrow +\infty$; so it attains the value $1/2$ exactly once there as well. Thus equation (21) has exactly two roots for $s = +1$. They are separated by $1/10$, because $C_+(1/10) < 1/2$ while $C_+(0) > 1/2$ and $C_+(1/4) > 1/2$; hence one root lies in $(0, 1/10)$ and the other in $(1/10, 1/4)$. Both are simple, since C'_+ vanishes only at ρ_* , where the value of C_+ is strictly below $1/2$.

For $s = -1$, admissibility is exactly $0 < \rho < 1/15$, and $C_-(\rho) = C_+(-\rho)$. On this interval,

$$\Xi(-\rho) < \frac{139}{9} - 191 < 0.$$

Indeed, the positive terms in $\Xi(-\rho)$ are bounded above by $22500/15^4 + 3375/15^2 = 139/9$. In equation (22), evaluated at $-\rho$, the factors $-\rho$ and $\Xi(-\rho)$ are both negative and every other factor is positive. Thus $C'_+(-\rho) > 0$, and consequently $C'_-(\rho) = -C'_+(-\rho) < 0$. Moreover,

$$C_-(0) = \frac{16}{27} > \frac{1}{2}, \quad \lim_{\rho \uparrow 1/15} C_-(\rho) = 0.$$

It follows that equation (21) has exactly one simple root for $s = -1$.

Each root with $s = +1$ supplies three angles, and the root with $s = -1$ supplies three more, giving nine off-axis points. Together with the two points in equation (19), this gives 11 critical points. At an off-axis critical point Lemma 5 applies, and the angular eigenvalue is

$$\rho^{-2} \partial_{\theta\theta} F = -90s\rho \neq 0.$$

Computing the meridional block of the Hessian and then imposing the two critical point relations (20) and (21) gives

$$\det D_{\rho,z}^2 F = \frac{-81\rho(2+10s\rho)^4}{(900\rho^4 + 360s\rho^3 + 36\rho^2 + 16)^{3/2}} C'_s(\rho). \quad (24)$$

This identity is asserted, and used, only at points where $C_s(\rho) = 1/2$. Its denominator is positive on the admissible intervals, as are ρ and $(2+10s\rho)^4$. Since all three roots of equation (21) are simple, $C'_s(\rho) \neq 0$; hence the meridional determinant is nonzero, with sign opposite to that of $C'_s(\rho)$. All nine off-axis points are therefore nondegenerate.

Finally we read off the Morse indices from Lemma 6, using the factorization $\det D^2 F = \rho^{-2} \partial_{\theta\theta} F \cdot \det D_{\rho,z}^2 F$ of Lemma 5. At the root with $s = +1$ lying in $(0, \rho_*)$ one has $C'_+ < 0$, so the meridional determinant is positive while the angular eigenvalue -90ρ is negative; the full determinant is negative and the index is 1. At the root with $s = +1$ lying in (ρ_*, ∞) one has $C'_+ > 0$, so both factors are negative and the index is 2. At the root with $s = -1$ one has $C'_- < 0$, so the meridional determinant is positive while the angular eigenvalue $+90\rho$ is positive; the index is 2. Each root contributes three points. Adding the two axial points, one of each index, gives

$$m_1 = 1 + 3 = 4, \quad m_2 = 1 + 3 + 3 = 7, \quad (25)$$

which completes the proof. $\square$

The 11 critical points in Lemma 8 form a finite set disjoint from e_z . Choose disjoint compact neighborhoods of these points, still avoiding e_z . Equation (17) gives C^2 -convergence on their union. The implicit-function theorem therefore gives 11 distinct nondegenerate critical points of K_γ , of the form $X = dY + o(d)$, for every sufficiently small $\gamma > 0$.

7 The remaining 17 points and the total count

Write

$$S_\gamma(X) = \frac{(2048/27)\gamma^4}{|X - de_z|}, \quad d = \frac{16}{3}\gamma,$$

so that $K_\gamma = P_\gamma + S_\gamma$. For $j = 0, 1, 2$, differentiation of the Newtonian kernel gives

$$|D^j S_\gamma(X)| \leq C_j \gamma^4 |X - de_z|^{-1-j}.$$

Call the four critical points of P_γ lying in an $O(\gamma)$ -neighborhood of the origin the *inner* ones, and the remaining 17 the *outer* ones; the latter comprise the eight of norm $\Theta(\sqrt{\gamma})$ and the nine with nonzero limits. This section shows that all 17 outer points persist in K_γ .

The persistence of the eight points of norm $\Theta(\sqrt{\gamma})$ is the delicate case, because their Hessians degenerate at rate γ while their mutual separations are only $\Theta(\gamma)$. We therefore record the quantitative form of the implicit function theorem that will be used.

Lemma 9 (Quantitative implicit function theorem). *Let $\overline{B}(x_0, r) \subset \mathbb{R}^m$ and let G be a C^1 map from $\overline{B}(x_0, r)$ to $\mathbb{R}^m$. Suppose that $L := DG(x_0)$ is invertible, put $\nu := \|L^{-1}\|^{-1} > 0$, and assume*

$$\|DG(x) - L\| \leq \frac{\nu}{2} \quad \text{for all } x \in \overline{B}(x_0, r), \quad \text{and} \quad |G(x_0)| \leq \frac{\nu r}{2}.$$

Then G has exactly one zero x_ in $\overline{B}(x_0, r)$, it satisfies $|x_* - x_0| \leq 2|G(x_0)|/\nu$, and $DG(x_*)$ is invertible.*

Proof. Put $\Phi(x) = x - L^{-1}G(x)$, so that $D\Phi(x) = L^{-1}(L - DG(x))$ and hence $\|D\Phi\| \leq \nu^{-1} \cdot \nu/2 = 1/2$ on the ball. Thus Φ is a $\frac{1}{2}$ -contraction there. Moreover $|\Phi(x_0) - x_0| = |L^{-1}G(x_0)| \leq |G(x_0)|/\nu \leq r/2$, so for $x \in \overline{B}(x_0, r)$,

$$|\Phi(x) - x_0| \leq |\Phi(x) - \Phi(x_0)| + |\Phi(x_0) - x_0| \leq \frac{1}{2}|x - x_0| + \frac{r}{2} \leq r.$$

Hence Φ maps the ball into itself and the Banach fixed point theorem supplies a unique fixed point x_* ; as L is invertible, the fixed points of Φ are exactly the zeros of G . The contraction estimate gives $|x_* - x_0| \leq (1 - \frac{1}{2})^{-1}|\Phi(x_0) - x_0| \leq 2|G(x_0)|/\nu$. Finally $\|DG(x_*) - L\| \leq \nu/2$ forces $DG(x_*)$ to be invertible. $\square$

Consider now the eight critical points of P_γ on the $O(\sqrt{\gamma})$ scale. At the two axial points, the eigenvalues divided by γ are $(-4, -4, 8)$. At each of the six off-planar points with radius $\rho_{o,-}$, the angular eigenvalue and the meridional Hessian, again divided by γ , have the limits

$$\gamma^{-1}\lambda_\theta \longrightarrow -12, \quad \gamma^{-1}D_{\rho,z}^2 P_\gamma \longrightarrow \begin{pmatrix} 4 & 0 \\ 0 & 8 \end{pmatrix}.$$

These limits follow from $\rho_{o,-} = 4\gamma/(3B) + O(\gamma^2)$ and $z^2 = \gamma/(2A) + O(\gamma^2)$. Since an orthonormal cylindrical frame is an orthogonal change of basis, the eight Cartesian Hessians divided by γ therefore have all eigenvalues bounded away from zero, uniformly for small γ ; fix $\nu_0 > 0$ with $\|(\gamma^{-1}D^2 P_\gamma(X_0))^{-1}\|^{-1} \geq \nu_0$ at each of the eight points X_0 .

By the last paragraph of the proof of Proposition 7, the mutual distances between these eight points are $\Theta(\gamma)$ — not $\Theta(\sqrt{\gamma})$, even though their norms are — because the six off-planar points lie at horizontal distance only $\rho_{o,-} = \Theta(\gamma)$ from the axis. Consequently they admit mutually disjoint closed balls of radius $c\gamma$ once $c > 0$ is a fixed sufficiently small constant. On such a ball,

$$\|D^2 P_\gamma(X) - D^2 P_\gamma(X_0)\| \leq c\gamma \sup |D^3 P_\gamma| = O(c\gamma),$$

because $D^3 P_\gamma = B D^3 H_3 + (A/4)D^3 H_4$, in which $D^3 H_3$ is a constant tensor and $D^3 H_4$ is linear in X ; on the balls this is bounded by an absolute multiple of $B + A\sqrt{\gamma} = O(1)$. Shrinking c therefore makes $\|D^2 P_\gamma(X) - D^2 P_\gamma(X_0)\| \leq \nu_0\gamma/4$ throughout the ball.

On these balls, $|X - de_z| \asymp \sqrt{\gamma}$, since $d = \Theta(\gamma)$. Consequently,

$$|\nabla S_\gamma| = O(\gamma^3), \quad |D^2 S_\gamma| = O(\gamma^{5/2}) = o(\gamma).$$

We apply Lemma 9 with $G = \nabla K_\gamma$, $m = 3$, $r = c\gamma$, and $L = D^2 P_\gamma(X_0)$, so that $\nu \geq \nu_0\gamma$. The first hypothesis holds because

$$\|D^2 K_\gamma(X) - L\| \leq \|D^2 P_\gamma(X) - D^2 P_\gamma(X_0)\| + \|D^2 S_\gamma(X)\| \leq \frac{\nu_0\gamma}{4} + O(\gamma^{5/2}) \leq \frac{\nu_0\gamma}{2}$$

for all small γ . The second holds because $\nabla P_\gamma(X_0) = 0$ gives $|G(X_0)| = |\nabla S_\gamma(X_0)| = O(\gamma^3)$, which is at most $\nu_0\gamma \cdot c\gamma/2$ for small γ . The lemma therefore yields exactly one critical point of K_γ in each ball, it is nondegenerate, and its displacement from X_0 is $O(\gamma^{-1}\gamma^3) = O(\gamma^2) = o(\gamma)$. As the balls are disjoint, the eight continuations are distinct.

The remaining nine outer critical points of P_γ converge to distinct nonzero limits. Their Hessians have invertible limits: this follows for the three planar points from the three eigenvalues displayed in Section 5, and for the six off-planar points from equation (15) and the nonzero angular eigenvalue. On fixed disjoint neighborhoods of these points, $\|S_\gamma\| * C^2 = O(\gamma^4)$. The implicit-function theorem therefore preserves all nine. We have proved that all 17 outer critical points of $P * \gamma$ persist in K_γ , with unchanged Morse indices, since a nondegenerate Hessian cannot change signature under a perturbation that keeps it invertible.

The 11 inner critical points found in Section 6 satisfy $|X| = O(d) = O(\gamma)$, whereas the preceding 17 points have distance $\asymp \sqrt{\gamma}$ or $\asymp 1$ from the origin. Thus the two families are disjoint for sufficiently small γ .

Thus K_γ has at least

$$17 + 11 = 28 \tag{26}$$

nondegenerate critical points for sufficiently small $\gamma > 0$. By the C^2 -convergence in Proposition 4, choose disjoint compact neighborhoods of these points that avoid de_z . For fixed γ , the implicit-function theorem gives one nondegenerate critical point of $\tilde{V}_{\gamma,\eta}$, and hence of $V_{\gamma,\eta}$, in each neighborhood for all sufficiently small $\eta > 0$. Since the corresponding X -coordinates remain bounded, all 28 lie at distance $O(\eta^2)$ from the origin in the original variables.

The 23 specified nonzero critical points of W have positive distance from the origin. Choose mutually disjoint compact neighborhoods of them that avoid the five original charge positions and the origin. On their union, the potentials of the second axial pair and the eighth charge tend to zero in C^2 , because their strengths are respectively $O(\eta^3)$ and $O(\eta^8)$, while their positions tend to the origin. All 23 critical points therefore persist as nondegenerate critical points of $V_{\gamma,\eta}$. They remain a fixed positive distance from the origin and are consequently disjoint from the 28 local points. Thus $V_{\gamma,\eta}$ has at least

$$23 + 28 = 51. \tag{27}$$

For comparison, if the eighth charge is omitted, all 21 critical points of P_γ persist and the same argument gives $23 + 21 = 44$ specified critical points for seven charges.

Remark 10 (Bookkeeping). It is suggestive to describe the eighth charge as replacing the four inner $O(\gamma)$ -scale points of P_γ by 11 points, so that

$$44 - 4 + 11 = 51.$$

This is a mnemonic only. What is proved above is the inequality $23 + 17 + 11 \leq N$, that is, that the 23 seed points, the 17 outer points, and the 11 innermost points are 51 distinct nondegenerate critical points. Nothing in the argument asserts that the four inner points of P_γ fail to persist; were any of them to persist, the count would only increase.

8 Removing all remaining degeneracies

The 51 counted points are already nondegenerate and therefore isolated, but the preceding argument does not exclude additional degenerate critical points. We now remove that possibility while preserving the 51 points. Keep the eight positions fixed, denote them by $b_1, \dots, b_8$, and vary the strength vector $q = (q_1, \dots, q_8)$ in the positive orthant.

On $M = \mathbb{R}^3 \setminus \{b_1, \dots, b_8\}$, consider the smooth family of electric fields

$$E(q, x) = \sum_{i=1}^8 q_i \frac{x - b_i}{|x - b_i|^3} = -\nabla V_q(x).$$

For fixed $x \in M$, the derivatives with respect to the strengths are

$$\partial_{q_i} E(q, x) = \frac{x - b_i}{|x - b_i|^3}.$$

These vectors span $\mathbb{R}^3$. Indeed, if a nonzero vector v were orthogonal to all of them, then $v \cdot b_i = v \cdot x$ for every i , so all eight positions would lie in one affine plane. This is impossible:

b_1, b_2, b_3 are the noncollinear triangle vertices in $z = 0$, while an axial charge lies outside that plane. Hence the map $E : \mathbb{R}^8 \times M \rightarrow \mathbb{R}^3$ is a submersion.

The positive orthant is open, so the parametric transversality theorem [9, p. 68], applied with the parameter restricted to this orthant, implies that arbitrarily close to the original strength vector there is a positive vector q' for which 0 is a regular value of $E(q', \cdot)$. Equivalently, every critical point of $V_{q'}$ is nondegenerate. Taking q' sufficiently close also preserves the 51 previously constructed nondegenerate critical points by the implicit-function theorem.

Every critical point lies in the convex hull of the charge positions. At a zero of E , one has

$$x = \frac{\sum_i (q'_i / |x - b_i|^3) b_i}{\sum_i q'_i / |x - b_i|^3},$$

which is a convex combination because all $q'_i > 0$. Near b_i , the term $q'_i(x - b_i)/|x - b_i|^3$ has norm $q'_i/|x - b_i|^2$, while the other seven terms remain bounded. Since the fixed strengths $q' * i$ are positive, a sufficiently small neighborhood of each charge contains no zero. The critical set is consequently a closed subset of the convex hull with these source neighborhoods removed, hence compact. Since every critical point is nondegenerate, it is also discrete; a compact discrete set is finite. The potential $V * q'$ is therefore Morse with a finite number of critical points, at least 51. This completes the proof of Theorem 1.

9 The Morse-index ledger

Every construction above produces critical points together with their Morse indices, and these indices satisfy a global constraint. We record it, both as an independent check on the count of Theorem 1 and because it localizes what any further critical points would have to look like.

Proposition 11 (Index identity). *Let $V = \sum_{i=1}^n q_i |x - b_i|^{-1}$ with all $q_i > 0$ and all b_i distinct, and suppose V is Morse. Let m_1 and m_2 denote the numbers of critical points of Morse index 1 and 2. Then every critical point has index 1 or 2, both numbers are finite, and*

$$m_2 - m_1 = n - 1. \quad (28)$$

Proof. Finiteness and the absence of indices 0 and 3 were established in Section 8 and Lemma 6, the latter applying because V is harmonic off the charges.

Put $E = -\nabla V$. Choose R so large that $\overline{B}_R(0)$ contains all charges and all critical points in its interior, and choose $r > 0$ so small that the closed balls $\overline{B}_r(b_i)$ are disjoint, contained in $B_R(0)$, and free of critical points. On the compact region $\Omega = \overline{B}_R(0) \setminus \bigcup_i B_r(b_i)$ the field E has no zeros on the boundary, so the Poincaré–Hopf index theorem gives

$$\sum_{p: E(p)=0} \text{ind}(E, p) = \deg\left(\frac{E}{|E|} \Big|_* \partial B_R(0)\right) - \sum_i \text{deg}\left(\frac{E}{|E|} \Big|_{\partial B_r(b_i)}\right),$$

the inner boundaries entering with the opposite sign. By the far-field estimate of Section ??, $E(x) = (\sum_i q_i)x|x|^{-3} + O(|x|^{-3})$, so for large R the field is homotopic through nowhere-vanishing fields to the outward radial field on $\partial B_R(0)$, and that degree is 1. Likewise $E(x) = q_i(x - b_i)|x - b_i|^{-3} + O(1)$ near b_i , whose first term has norm $q_i/r^2 \rightarrow \infty$, so for small r each inner degree is also 1. Hence the index sum equals $1 - n$. (Both far-field statements follow from the mean-value theorem applied to $u \mapsto u|u|^{-3}$, as in Section ??.)

At a nondegenerate zero p of E of Morse index k ,

$$\text{ind}(E, p) = \text{sign det } DE(p) = \text{sign det } (-D^2V(p)) = (-1)^3(-1)^k = (-1)^{k+1}.$$

Therefore $\sum_p (-1)^{k(p)+1} = 1 - n$, that is, $\sum_p (-1)^{k(p)} = n - 1$. Since $k \in \{1, 2\}$, the left-hand side is $m_2 - m_1$. $\square$

For $n = 2$ equation (28) says that the unique critical point is a 2-saddle. For the equilateral triangle of Section 2 it determines the indices outright: by the expansion $V_\Delta = 3 + \frac{3}{4}H_2 + \dots$ the center has Hessian $\frac{3}{4}\text{diag}(2, 2, -4)$ and hence index 1, so with $m_1 + m_2 = 4$ and $m_2 - m_1 = 2$ the three noncentral critical points all have index 2.

We now assemble the ledger for the configuration of Theorem 1. For this we take the 23 seed points to be those produced by the construction of [8], namely the three noncentral critical points of the triangle together with the 20 nonzero members of the 21-point family on the scale ϵ^2 . The proof of Proposition 7 uses only that the three coefficients are positive and satisfy the smallness condition (12), so the proposition applies verbatim to any polynomial $\gamma H_2 + B H_3 + \frac{A}{4} H_4$ with those properties. The 21-point family of [8] is the critical set of $\frac{5}{32} H_2 + \frac{15}{8} H_3 + \frac{3}{16} H_4$, that is, of the triple

$$A = \frac{3}{4}, \quad B = \frac{15}{8}, \quad \gamma = \frac{5}{32}, \quad \text{for which} \quad \frac{A\gamma}{B^2} = \frac{1}{30} < \frac{3}{80}.$$

Equation (16) therefore applies to it: it has $m_1 = 10$ and $m_2 = 11$. Discarding the origin, which has index 1, and adjoining the three noncentral triangle points, which have index 2, the 23 seed points have $m_1 = 9$ and $m_2 = 14$. For the 17 outer points we discard from equation (16) the origin (index 1) and the three inner planar points of radius $\rho_{p,-}$ (index 2), leaving $m_1 = 9$ and $m_2 = 8$. The 11 innermost points have $m_1 = 4$ and $m_2 = 7$ by equation (25). Morse indices are unchanged by all the continuations used above: along each of them the Hessian varies continuously and remains invertible, so no eigenvalue can cross zero and the signature is constant; and the intermediate rescalings multiply Hessians by positive constants, which also preserves signature. Hence:

Family	m_1	m_2	total
23 seed points (Theorem 2)	9	14	23
17 outer points (Section 7)	9	8	17
11 innermost points (Lemma 8)	4	7	11
Total	22	29	51

The ledger closes: $29 - 22 = 7 = 8 - 1$, in exact agreement with equation (28) for $n = 8$. The same computation for the seven-charge construction of Section 7 gives $m_1 = 9 + 10 = 19$ and $m_2 = 14 + 11 = 25$, and $25 - 19 = 6 = 7 - 1$; and for the five-charge seed it gives $m_1 = 10$, $m_2 = 14$, and $14 - 10 = 4 = 5 - 1$.

Two consequences deserve mention. First, the ledger is an independent check on the combinatorics of the construction, carried out with no reference to the counting arguments that produced it. Second, by Proposition 11 the 51 constructed points already exhaust the index identity, so any further critical point of $V_{q'}$ must be matched by another of the opposite index. The construction is therefore consistent with $N = 51$ exactly, although we do not prove that no further critical points exist.

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